GIRIDHARAN S

Computer Science Undergraduate · Robotics · AI/ML · Software Engineering

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SUMMARY

Computer Science undergraduate at VIT Vellore who builds intelligent software and robotics systems: ROS 2 mapping and navigation, AI/ML evaluation, explainable neuro-symbolic agents, computer vision and verifiable databases. Strong in Python, C/C++ and Java. Seeking robotics, AI/ML or software engineering internships.

EDUCATION

B.Tech, Computer Science | Vellore Institute of Technology, Vellore | **CGPA: 9.19 (till date)** — **08/2024 – 05/2028**

Higher Secondary Education | Samsidh International School | **85%** — **04/2022 – 05/2024**

Secondary Education | Chinmaya Vidyalaya | **88%** — **04/2015 – 05/2022**

EXPERIENCE

Junior Software Engineer | **Team Artemis**, *Vellore Institute of Technology, Vellore* — **07/2026 – Present**

- Mapped the Team Artemis lab (physical LiDAR) and a Gazebo maze in ROS 2 using Hexapod and TurtleBot
- Ran 2 navigation tests to predefined goals on saved maps; learning Nav2 and exploring a modified approach
- Supporting EYIC and Technocian finals preparation; assisting juniors in RoboReach at Techfest, IIT Bombay

AI/ML Intern | **IITM Pravartak (IIT Madras)**, *Chennai, Hybrid* — **05/2025 – 06/2025**

- Validated ~800 Hindi speech-to-text test cases for Sansad TV broadcasts against ground-truth transcripts
- Evaluated model outputs for accuracy, consistency and edge-case handling, logging findings for model iteration

PROJECTS

CineTrace — Explainable & Adaptive Movie Recommender | *React, FastAPI, PostgreSQL, SQL, pytest* — **10/2026 – 11/2026**

- Designed an 18-table PostgreSQL schema linking each explanation by foreign keys to the exact rows that justify it
- Built trigger-maintained, append-only rating history and a time-travel SQL verifier: 0 of 4,294 claims invalid
- Enforced integrity with CHECKs, deferred and immutability triggers; drift-aware item-kNN with MMR; 40 pytest tests

Knowledge-Augmented Neuro-Symbolic Travel Planner | *Python, Flask, SQLite, scikit-learn* — **08/2026 – Present**

- Neuro-symbolic agent that turns natural-language requests into constrained flight × hotel plans, with risk checks
- Asks only if an answer changes the plan: 6 vs 9 first-trip questions at 100% oracle match; 158 tests pass

Mini Compiler for a Custom Language | *Python, unittest · team of 3* — **Course project**

- Seven phases: lexer, recursive-descent parser, semantics, TAC, optimizer, 8-register codegen, cycle-counting VM
- Cost-gated optimizer applies a rewrite only if program cost drops: 33.7% fewer VM cycles than standard passes

GreenPlace — SDN Controller Placement | *ILP, NSGA-II, Mininet, Ryu · group project* — **08/2026 – Present**

- Energy-aware placement via ILP and NSGA-II for static and dynamic traffic, evaluated on 46 Topology Zoo networks

Smart Traffic Management System | *Python, OpenCV, ESP32* — **04/2026 – 08/2026**

- OpenCV vehicle detection drives real-time signal decisions; rule-based Python logic controls hardware via ESP32

TRAINING & CERTIFICATIONS

Robotic Fundamentals and Introduction to ROS (RViz, Gazebo, SLAM) · VIT Value Added Course — **01/2026 – 04/2026**

Machine Learning & AI Fundamentals Training Program · TEACHNOOK, Remote — **10/2024 – 12/2024**

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL
Robotics: ROS 2, RViz, Gazebo, LiDAR, URDF, Nav2 (learning)
AI/ML & Vision: scikit-learn, OpenCV, model evaluation
Backend & Databases: FastAPI, Flask, PostgreSQL, SQLite
Frontend: HTML, CSS, JavaScript, React, Vite
Tools: Git, GitHub, VS Code, Jupyter, pytest, ESP32, Mininet
Concepts: DSA, OOP, DBMS, AI/ML, Computer Vision, Compiler Design
Problem Solving: 55+ LeetCode problems (Arrays, Strings, DP, Graphs)